

Chapter 5

Data-Driven Temporal Computing with Polymer-Electrolyte Memristor Reservoirs

5.1 Introduction and Scope

???? established and quantified the slow, volatile response of these polymer-electrolyte composites. Here, behavioural models represent the measured response and are evaluated on temporal-computing tasks. A reservoir maps a time-varying input onto a high-dimensional dynamical state from which a linear read-out can recover temporal information.

The central comparison is between a bank drawn from one composition and a bank spanning several measured timescales. The latter should help when a target depends on several lags, but need not improve a task dominated by one timescale. Wearable *affective computing* (the recognition of emotional and physiological state) provides a relevant

test because its signals contain both fast and slow components. The evaluation proceeds from reservoir benchmarks to physiological streams, WESAD classification, continuous valence/arousal tracking, and tests with sensor noise and wrist-band data.

All results in this chapter are *in silico*; no new devices were fabricated. Each node is a compact behavioural model parameterised by the fits in ?? (section 5.2), and every read-out is a single linear ridge regression. Demonstration A tests whether one composition suffices for a single-timescale feature. Demonstration B tests whether a composition-diverse bank improves multi-timescale tasks.

5.2 Measured Behavioural Model

Each composition cell is reduced to a compact behavioural model and used as one reservoir node. The model is fitted only to the three common measurements of ??: hysteresis, pulsed potentiation, and delayed depotentiation. Its parameters therefore derive from measured device data.

Each composition cell is summarised by a *parameter card*: the write nonlinearity ϕ_c from the potentiation curves, the fading-memory law λ_c from the depotentiation curves, and an associated device-to-device spread. The fading memory is the stretched-exponential (Kohlrausch) retention identified in ??,

$$\lambda_c(\Delta t) = \exp\left[-(\Delta t/\tau_c)^{\beta_c}\right],$$

where τ_c and β_c are the per-cell relaxation time and stretch exponent; for the few cells without an identified stretched fit, a single exponential reproducing the model-free half-life is used instead ($\tau_c = t_{1/2}/\ln 2$, $\beta_c = 1$). The write nonlinearity is the compressive, saturating build-up of conductance with pulse number measured in ??, parametrised by its

log-log growth exponent α_c and its peak enhancement.

Driven by a rate-coded input $u_n \in [0, 1]$ sampled at interval Δt , a single device node updates as

$$x_n^{(i)} = \lambda_i x_{n-1}^{(i)} + \left[\max(w_i u_n, 0) \right]^{\alpha_i}, \quad \lambda_i = \exp \left[- (\Delta t / \tau_i)^{\beta_i} \right],$$

so that the node integrates a compressive write of its masked input and then leaks according to the measured retention λ_i over the sample interval; w_i is the reservoir input mask. The internal state is read sub-threshold and is assumed to be state-preserving, as in the proof-of-concept device of ???. Every read-out is a single *linear* ridge regression on the device states.

The model has two measurement-related constraints. First, write nonlinearity and fading memory were measured at different write amplitudes (3 V for potentiation and 4 V for depotentiation; ???). Combining them in one node assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. Second, both measurements used a single, effectively common inter-pulse cadence, so the model does not resolve input timing below that cadence. The gold/TMPE negative-polarity corpus of ??? measures integration and relaxation within single runs and finds both interval-dependent potentiation and written-depth-dependent relaxation. Because it uses another host, electrode, polarity, and read protocol, it does not provide parameters for the banks used here. The experiment needed for the lead composition is specified in section 5.12. Per-cell parameter cards and all simulation and read-out settings are tabulated in Appendix D.

5.3 Reservoir Computing and the Common Metric Language

The reservoir principle. Reservoir computing uses a fixed dynamical system whose state is connected to a trained linear read-out [1–3]. A physical reservoir requires fading memory, so its state retains recent input history, and a nonlinear response that maps inputs into features accessible to the read-out. ?? measures both properties in these devices: stretched-exponential depotentiation supplies fading memory, and compressive potentiation supplies nonlinearity. Memristive and organic implementations have previously used the same combination [4–7]. This chapter evaluates the polymer-electrolyte devices as another implementation.

Architecture. The reservoir is a spatial bank of N *independent* device nodes, each represented by the behavioural model of section 5.2 and driven by a randomly masked copy of the input. With no inter-node recurrence, differences between banks arise from their intrinsic timescales, device-to-device scatter, and input masks rather than engineered connectivity. For Demonstration A, the same single-composition node could instead be deployed as a time-multiplexed delay-feedback reservoir with virtual nodes along a delay line [8]; the spatial bank is used throughout for consistency.

Common metrics. Linear memory capacity MC_k and its total [9] quantify the past input recoverable at each lag. NARMA-10 [10] provides a nonlinear system-identification benchmark, and the information-processing-capacity framework [11] separates linear and nonlinear contributions (fig. 5.3). Affective tasks are scored by class-balanced macro-F1. Robustness is tested by injecting the measured device-to-device and cycle-to-cycle spread into the bank (fig. 5.9).

5.4 Benchmark Tier: Capacity and System Identification

Reservoir suitability is first evaluated without an application-specific task. Linear memory capacity measures fading memory, while information-processing capacity separates its linear and nonlinear components. The NARMA-10 system-identification task is used only to rank compositions, not to claim absolute performance. All three benchmarks use a random rate-coded input, N device nodes, and a linear ridge read-out.

Statistical units. Random input and bank seeds sample the algorithm’s sensitivity to stochastic initialisation; they are not independent devices, experiments, or subjects. Seed-only benchmark comparisons therefore report means, standard deviations, and sign consistency without inferential p -values. Inferential paired tests are reserved for independent held-out subjects in the WESAD and CASE analyses, or for independently fabricated devices in the experimental chapters. The complete distinction is recorded in Appendix D.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total is the sum over all lags, bounded above by the number of nodes N [9]. Figure 5.1 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09, differing only by device-to-device scatter) and a *heterogeneous* bank spanning the composition grid (τ from ≈ 3 to ≈ 26 s). The heterogeneous bank broadens the memory profile and raises total capacity from 4.2 to 6.4 (means over ten random input/bank seeds), a factor of 1.54 ± 0.10 . Single-step recall rises from $MC_1 = 0.51$ to 1.00,

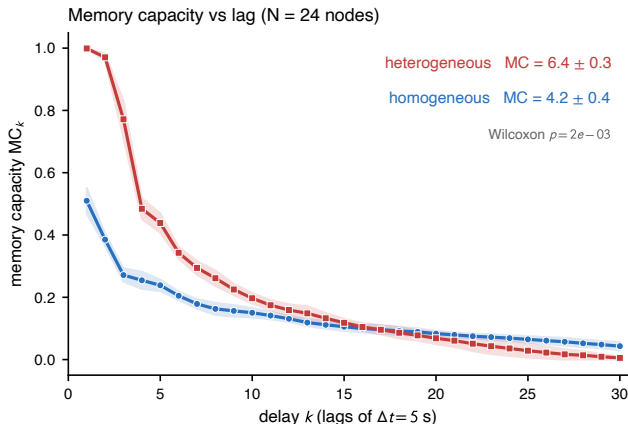


Figure 5.1: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 3\text{--}26\text{ s}$). The heterogeneous bank broadens recoverable memory across lags, raising total memory capacity from 4.2 to 6.4 (a $1.54\times$ increase, positive in all ten seed pairs), with the gain concentrated at short-to-intermediate lags. Shaded bands are ± 1 SD over ten random input/bank seeds. Seeds quantify algorithmic sensitivity rather than experimental replication. $N = 24$ leaky nodes, random uniform input, linear (ridge) read-out; in-silico devices built from the ?? parameter cards.

with most of the gain at short-to-intermediate lags. The increase occurs in all ten seed pairs; this is a sensitivity result, not a population-level significance test. Because the nodes are independent, the increase comes from their timescale distribution, device scatter, and input masks.

Composition sweep and the single-node optimum. The same benchmarks, run on single-composition banks, rank the compositions against one another and so test the Demonstration-A claim that one cell suffices for a single-timescale task. Figure 5.2 shows that the lead cell PEO 0.3/0.09 attains the highest single-composition memory capacity (3.98 ± 0.19 , mean $\pm$ SD over ten seeds), while the neighbouring PEO 0.3/0.045 cell is marginally better on the NARMA-10 system-identification benchmark [10] (NRMSE 0.587 ± 0.019 against the lead’s

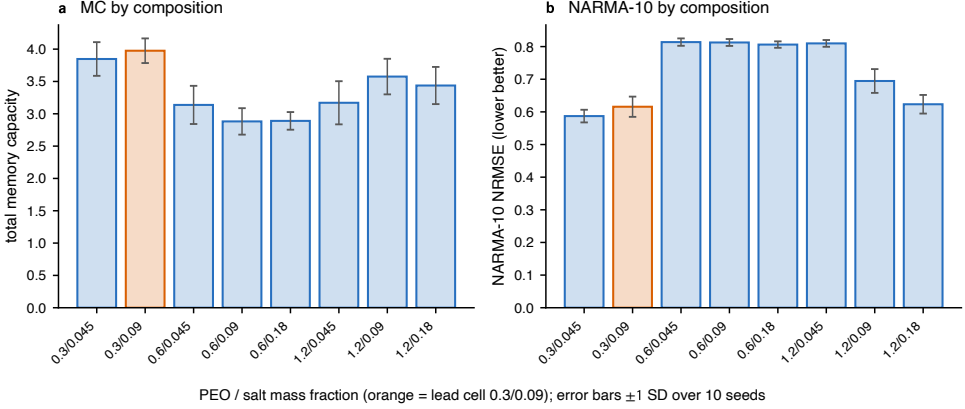


Figure 5.2: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The lead cell PEO 0.3/0.09 (orange) attains the highest memory capacity; the neighbouring 0.3/0.045 cell is marginally better on NARMA-10. Both optima fall in the low-PEO row anticipated by $??$. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

0.616 ± 0.031 , a close second). Both optima fall in the low-PEO row that $??$ identifies as carrying the longer fading-memory times, so the benchmark ranking is consistent with the composition result, not an artefact of the simulation. This supports the Demonstration-A statement: for a single-timescale feature, the lead composition is the memory-capacity optimum and a heterogeneous bank is unnecessary. The NARMA-10 numbers carry only *relative* weight: the absolute error is high (≈ 0.6 – 0.8) because a bank of independent, non-recurrent nodes is a weak NARMA-10 reservoir, so the stronger evidence here comes from the capacity measures above (figs. 5.1 and 5.3) and NARMA-10 serves only to corroborate the composition ranking, a limitation a recurrent or delay-feedback architecture would relax.

Comparison with randomly distributed timescales. A generic echo-state bank of the same size tests whether the measured timescales are individually important or whether their range is sufficient. A random bank spanning the same range as the compositions ($\tau \approx 3\text{--}26\text{ s}$) has a memory capacity of 6.7 ± 0.3 , compared with 6.4 ± 0.3 for the device bank (ten matched seeds; mean difference $+0.28$). Expanding the random range to $0.5\text{--}60\text{ s}$ raises the capacity to 7.6 ± 0.3 (mean difference $+1.14$, positive in all ten seed pairs). The measured τ values are therefore not uniquely optimal; the relevant property is the range of timescales. The broader-range control also indicates that extending this range, for example through the drive-amplitude dependence measured in ??, could increase capacity further.

Linear and nonlinear capacity. Memory capacity counts only linearly recoverable information. The information-processing-capacity framework of Dambre et al. decomposes total capacity into orthogonal polynomial degrees: degree-one terms give linear capacity, while degree-two terms measure nonlinear self and cross-lag transformations; the total is bounded by the node count N [11]. Figure 5.3 reports this decomposition. The homogeneous bank has a nonlinear capacity of 2.7 within a total of 7.1, showing that compressive potentiation contributes a nonlinear response in addition to fading memory. The heterogeneous bank raises the mean total capacity to 11.6, with the increase present in all ten seed pairs and in both components. Almost all nonlinear capacity comes from degree-two self terms; cross-lag products are negligible, as expected for an independent, non-recurrent bank. The total remains below the N bound for both banks.

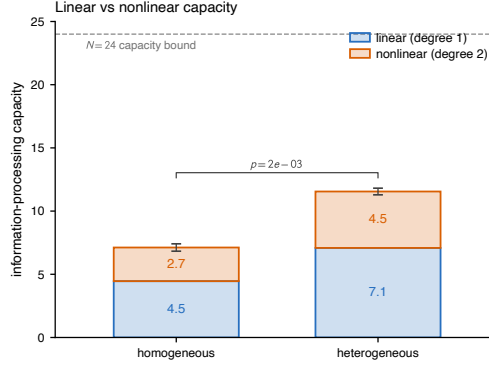


Figure 5.3: Information-processing capacity [11] split into linear (degree-one) and nonlinear (degree-two) parts for homogeneous and heterogeneous banks ($N = 24$, ten seeds). Both banks have nonzero degree-two capacity, indicating a nonlinear contribution from the compressive write. Heterogeneity raises mean total capacity from 7.1 to 11.6, with a positive difference in all ten seed pairs. The dashed line is the $N = 24$ upper bound; error bars are ± 1 SD on the total over seeds, which quantify algorithmic sensitivity.

5.5 Affective Computing

Affective computing uses physiological signals to infer emotional and physiological state [12]. The device timescales, from a few seconds to tens of seconds, overlap the phasic electrodermal, respiration, and heart-rate-variability bands without time rescaling (table 5.1). These signals also contain several timescales at once: phasic electrodermal responses, respiration and heart-rate-variability rhythms over seconds, and tonic levels drifting over minutes [13–15]. Figure 5.4 places the measured relaxation time of each composition on the same axis as these physiological bands and shows the minutes-scale range not reached by composition alone.

Extending the model to tonic timescales. Raising the write amplitude lengthens fading memory in the measurements of $??$. Adding modelled, drive-boosted minutes-scale nodes nearly doubles recall beyond

Table 5.1: Characteristic timescales of the peripheral affective signals used in this chapter, and the composition cells whose fading-memory window covers them [13–15]. The grid of ?? spans the full range, with the lead PEO 0.3 cells serving the slow LF-HRV band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	$\sim 1\text{--}10\text{ s}$	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	$\sim 3\text{--}5\text{ s}$	mid PEO
HRV, HF band	$\sim 2.5\text{--}7\text{ s}$	mid PEO
HRV, LF band	$\sim 7\text{--}25\text{ s}$	PEO 0.3 ($\tau \approx 19\text{--}26\text{ s}$)
Tonic SCL	minutes	slowest / drive-boosted

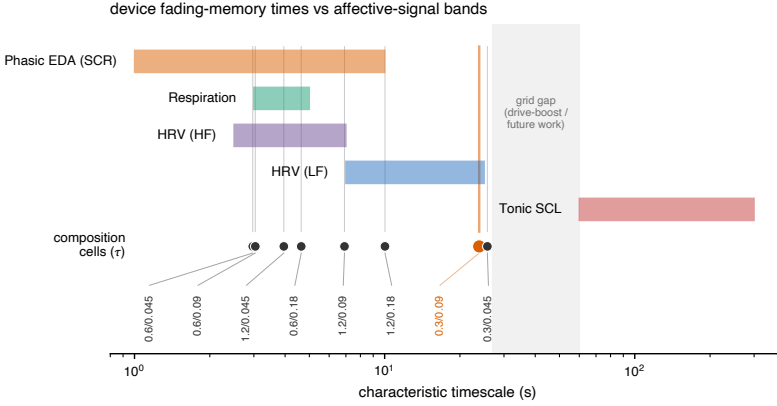


Figure 5.4: Each PEO/LiOTf composition cell is placed at its measured fading-memory time τ (markers, lower track) on the same logarithmic axis as characteristic timescale bands of peripheral affective signals (coloured bars; table 5.1). The composition grid ($\tau \approx 3\text{--}26\text{ s}$) overlaps the phasic-EDA, respiration, and HRV bands, with the lead cell PEO 0.3/0.09 (orange) in the slow LF-HRV band. Composition alone does not reach the tonic, minutes-scale band (shaded gap); testing that range would require drive-amplitude measurements or a slower cell.

75 s and raises total memory capacity from 6.4 to 7.0. On the physiological streams, reconstruction at delays up to 120 s improves by +0.019 in held-out R^2 , with a positive difference in all ten seed pairs (Appendix D). This is a modelled extrapolation because no minutes-scale cell was measured. A drive-amplitude experiment at these timescales is required to test it.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [16]: fifteen subjects wearing a chest sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways: as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 5.6), and as a downstream affective-classification task (section 5.7).

5.6 Physiological Temporal-Context Reconstruction

The benchmarks of section 5.4 show that a spread of device timescales broadens memory on a random input. This section tests the same property on the multi-timescale physiological signals introduced in section 5.5, isolating temporal-context encoding before turning to affect labels (section 5.7).

The reservoir is run continuously over each WESAD session at $\Delta t = 1$ s on four chest channels (electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate), and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [13, 14]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Table 5.2 reports the held-out reconstruction quality. A memoryless bank gives $R^2 = 0.674$, compared with 0.673 for instantaneous input. A single-timescale memory raises R^2 to about 0.74, and the heterogeneous composition bank reaches 0.756: +0.084 over instantaneous input and

Table 5.2: Leave-one-subject-out held-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. Fast and slow homogeneous banks give their highest scores in their corresponding bands. Across the full context vector, the heterogeneous bank exceeds the best homogeneous bank by +0.012 R^2 , with a positive difference in all fifteen subjects (paired Wilcoxon $p < 10^{-4}$).

Bank	overall R^2	fast	mid	slow
Instantaneous	0.673	0.715	0.670	0.632
Memoryless	0.674	0.721	0.667	0.630
Homogeneous, fast	0.741	0.815	0.735	0.669
Homogeneous, slow	0.744	0.787	0.724	0.711
Heterogeneous	0.756	0.815	0.737	0.708

+0.012 over the best same-size homogeneous control. This difference is positive for all fifteen subjects (paired Wilcoxon signed-rank $p < 10^{-4}$). In the band-resolved results, the fast homogeneous bank performs best at the shortest delays and the slow homogeneous bank at the longest. The heterogeneous bank gives the highest single-bank score across the full delay range (fig. 5.5).

5.7 WESAD Affective Classification

The downstream task is affective-state classification from the WESAD physiological streams with a leave-one-subject-out linear read-out. The analysis reports the contribution of timescale heterogeneity with uncertainty, including when the increment does not differ from zero. The study that introduced WESAD reports three-class chest macro-F1 ≈ 0.72 with engineered window features and conventional classifiers under the same leave-one-subject-out protocol [16]; the present streaming model reaches 0.758 under that matched metric and subject split (table 5.3). Accuracy-only results or studies using different validation splits are excluded from

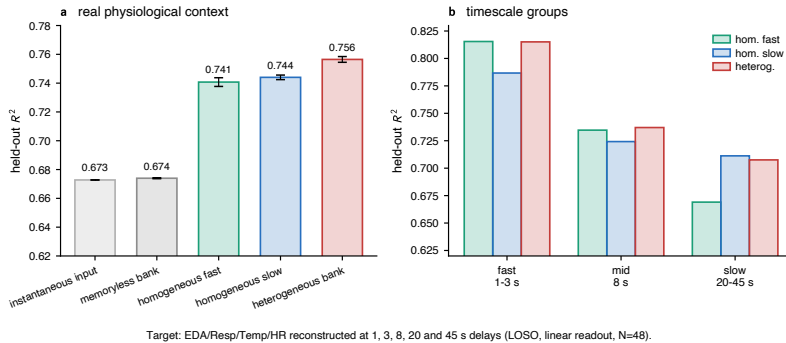


Figure 5.5: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and ECG-derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45s from the reservoir state (leave-one-subject-out, $N = 48$). The heterogeneous composition bank gives the best single-bank held-out R^2 (0.756) across the full fast-to-slow context vector, ahead of the best same-size homogeneous control (0.744) and well ahead of instantaneous input (0.673). Fast and slow homogeneous banks each win only their own band.

Table 5.3: Matched comparison on three-class WESAD chest signals. Both rows report class-balanced macro-F1 under leave-one-subject-out validation over the same fifteen subjects. The models and temporal formulations still differ, so the table provides a scale reference rather than a superiority claim. Accuracy-only and differently split studies are omitted.

Study	Features and read-out	Validation	macro-F1
Schmidt et al. (2018) [16]	engineered window features, AdaBoost/RF	15-subject LOSO	≈ 0.72
This work	streaming device-model states, linear ridge read-out	15-subject LOSO	0.758

this numerical comparison because they estimate a different quantity.

Demonstration A: window-level electrodermal classification. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches a leave-one-subject-out macro-F1 of 0.894 (fig. 5.6a). A static baseline using

the mean, standard deviation, last value, and slope of the same signal reaches 0.899. The 60-second-window task therefore does not demonstrate a contribution from temporal memory or timescale heterogeneity. This motivates the streaming formulation below.

Demonstration B: streaming three-class tracking. The reservoir runs continuously over each session, and the affective class (baseline, stress, or amusement) is read out at every step. This formulation uses the evolving reservoir state without a static window summary. Table 5.4 decomposes the result into four matched-size banks. Under the same leave-one-subject-out protocol, a majority-class predictor reaches macro-F1 0.23 and a stratified-random predictor 0.33, compared with 0.758 for the heterogeneous reservoir. Bank comparisons are assessed across the fifteen independent held-out subjects, preventing within-subject temporal autocorrelation from inflating the test. Relative to instantaneous input, a same-size memoryless bank contributes +0.028, a homogeneous fading-memory bank adds +0.016, and timescale heterogeneity adds +0.005. The fading-memory gain over a same-size memoryless bank is +0.014 to +0.020. The heterogeneity increment is $+0.005 \pm 0.011$, within seed variation; it is positive in seven of ten seeds, and remains in $[+0.002, +0.007]$ across sampling intervals from 0.5 to 4s without separating from zero. Per-class F1 is 0.90 for baseline, 0.92 for stress, and 0.55 for amusement. Amusement has the least labelled data (roughly 5.6 ks, compared with 17.6 ks for baseline and 10.0 ks for stress) and the lowest autonomic arousal. Its score remains low across all banks, so this class does not distinguish memory configurations.

Why heterogeneity does not improve the label scores. The affect labels are slowly varying sustained states, and their discriminative memory is dominated by the slow tonic band already covered by the

Table 5.4: Leave-one-subject-out macro-F1 for streaming three-class WESAD tracking (means $\pm$ SD over ten bank seeds; matched input masks). Each increment is the gain over the row above: fading memory contributes a small, consistent gain, while the timescale-heterogeneity increment is within noise (*ns*).

Bank	macro-F1	increment
Instantaneous (no memory)	0.709	—
Memoryless (dimensionality only)	0.737	+0.028
Homogeneous (memory, single τ)	0.753 ± 0.009	+0.016
Heterogeneous (memory, τ spread)	0.758 ± 0.006	+0.005 (<i>ns</i>)

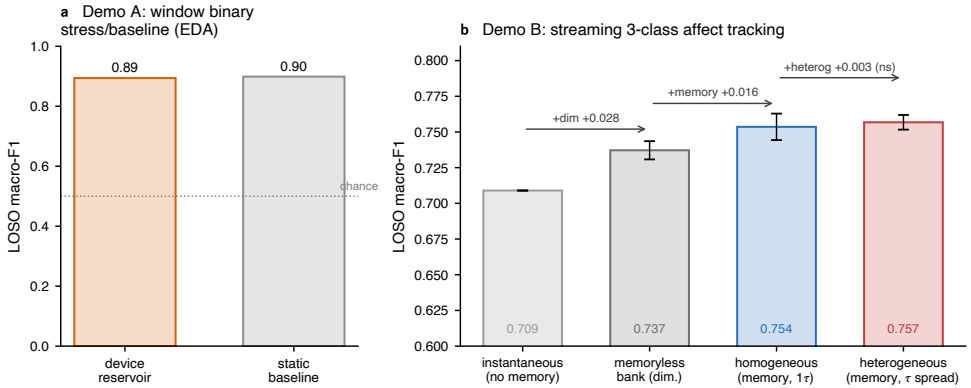


Figure 5.6: Affective classification with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Demonstration A, window-level binary stress-versus-baseline from EDA: the single lead composition matches a static feature baseline, i.e. the 60 s-window task is quasi-static and requires no fading memory. (b) Demonstration B, streaming three-class affect tracking (baseline/stress/amusement) decomposed as instantaneous $\rightarrow$ +dimensionality $\rightarrow$ +fading memory $\rightarrow$ +timescale heterogeneity. Fading memory contributes a small but consistent gain; the timescale-heterogeneity increment ($+0.005 \pm 0.011$) is within seed noise. Error bars are ± 1 SD over 10 bank seeds.

lead composition ($\tau \approx 19\text{--}26$ s). A wider timescale distribution improves performance when a target requires several lags simultaneously, as in the memory-capacity and physiological-context tasks (fig. 5.1 and table 5.2). The sustained label task does not show that requirement. Under this pro-

tol, fading memory helps classification, but the additional heterogeneity increment is not resolved.

5.8 Continuous Valence and Arousal

The WESAD tasks of section 5.7 use categorical labels dominated by sustained states. Continuous valence (pleasantness) and arousal (activation) provide a dimensional target with fast phasic components and slow tonic drift [12]. This third corpus tests whether a spread of timescales improves prediction of that continuous target.

The Continuously Annotated Signals of Emotion corpus (CASE) of Sharma et al. [17] records thirty subjects who rated valence and arousal continuously on a joystick (range [0.5, 9.5], 20 Hz) while watching emotional videos, with synchronised physiology at 1 kHz. The reservoir is run over the eight emotional videos at $\Delta t = 1$ s on the same slow channel set as WESAD (electrodermal activity, respiration, skin temperature, and an electrocardiogram-derived heart rate), with robust scaling per subject. The state is reset and a short washout is removed at each video boundary to prevent persistence across stimuli. A linear ridge read-out predicts valence and arousal under leave-one-subject-out evaluation, scored by the concordance correlation coefficient (CCC) [18].

Every fading-memory bank outperforms the instantaneous read on the CASE task (the heterogeneous bank by +0.079 mean CCC, positive in 24 of 30 subjects, paired Wilcoxon $p < 10^{-4}$). A multi- τ filter bank raises CCC from 0.23 for a single timescale to 0.41 (+0.176, 27/30 subjects, $p < 10^{-6}$). Timescale spread therefore improves this continuously annotated target, consistent with the memory-capacity and physiological-context results of sections 5.4 and 5.6 and unlike the sustained WESAD labels.

On this smooth regression, a linear multi- τ filter uses the timescale spread

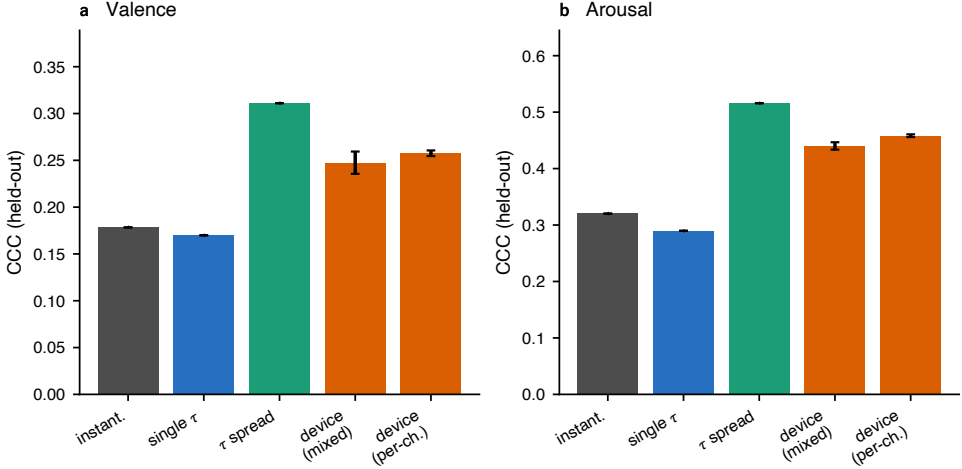


Figure 5.7: Continuous affect tracking on the CASE corpus [17] (thirty subjects; held-out concordance correlation coefficient, leave-one-subject-out, linear read-out; mean $\pm$ SD over ten bank seeds). (a) Valence and (b) arousal. A linear multi- τ filter bank increases CCC by +0.176 over a single- τ filter, with a positive difference in 27/30 subjects ($p < 10^{-6}$). Device banks with mixed and per-channel input masks also use a spread of timescales but score below the linear filter on this regression task.

more effectively than the device model. A per-channel device bank reaches CCC 0.358, compared with 0.413 for the linear bank (-0.047 , $p < 10^{-2}$ across the thirty held-out subjects); random cross-channel mixing in the standard bank reduces the score by a further ≈ 0.014 . The device models therefore do not outperform an equivalent digital filter on this task. Their proposed advantage is passive material-level filtering at the core-power estimate of section 5.10. The increment from device-internal heterogeneity is not resolved across held-out subjects ($+0.005$, $p = 0.18$; subject values are seed-averaged), because the jittered homogeneous bank already contains a timescale spread (Appendix D).

5.9 Continuous Monitoring Under Deployment Conditions

The sustained WESAD states can often be classified from the instantaneous signal level, as shown by the static-feature baseline. Wearable measurements also contain sensor noise and motion artefacts, particularly in wrist-worn electrodermal and photoplethysmographic signals. Fading memory implements temporal low-pass filtering and may therefore reduce sensitivity to transient artefacts. The following experiments test that effect directly.

Figure 5.8 tests a continuous binary stress detector (stress versus baseline plus amusement, read out at every step, leave-one-subject-out, with the same linear read-out used throughout). On clean streams, all banks reach macro-F1 ≈ 0.91 – 0.92 , comparable to the $\approx 93\%$ binary benchmark for the corpus [16]. With increasing sensor noise and motion-artefact bursts, the instantaneous and memoryless controls fall from 0.914 to 0.840 at $\sigma = 0.4$ of the signal range, while the fading-memory banks remain near 0.92. At $\sigma = 0.4$, the heterogeneous bank exceeds the instantaneous read by +0.088 for all fifteen subjects (paired Wilcoxon $p < 10^{-4}$). On clean streams, fading memory reduces the false-alarm rate from 0.046 to 0.026. Homogeneous and heterogeneous banks have similar robustness, associating the result with their shared fading memory. Near label transitions, the increase from 0.747 to 0.775 is not resolved across subjects, and detection latency is similar across banks because it is set by the prediction-smoothing horizon.

Accelerometer-gated motion artefact. Because the preceding corruption is synthetic, a second test gates its amplitude with the measured chest accelerometer. At each step, injected amplitude follows measured movement rather than being distributed uniformly. At the strongest

gating, the instantaneous read falls from macro-F1 0.914 to 0.898, while the memory bank remains near 0.92. The +0.032 difference is positive across subjects (paired $p = 0.04$), and a classical moving average gives a similar +0.036 difference (Appendix D). The effect is smaller than under the synthetic injection, consistent with a mechanically stable chest strap and a largely seated protocol. Wrist measurements are examined in section 5.10.

Classical-filter control. An exponential moving average applied to the same raw channels tests whether the noise robustness requires the device model. A single time constant matched to the lead cell reaches macro-F1 0.927 at $\sigma = 0.4$, compared with 0.840 for the instantaneous read and 0.916 for the device bank (+0.091 over instantaneous, all fifteen subjects, $p < 10^{-4}$). Spreading the moving-average timescales adds +0.005 (*ns*), and the device’s compressive nonlinearity adds -0.008 relative to that linear filter bank (*ns*; Appendix D). These controls attribute the robustness to fading-memory filtering; the device-specific nonlinearity provides no resolved increment. The proposed device contribution is to perform this filtering through material relaxation, with no trained dynamics and at the core-power estimate of section 5.10. The multi-timescale benchmarks of sections 5.4 and 5.6 distinguish the heterogeneous bank; this single-timescale stress detector does not.

Cross-corpus replication. The robustness test was repeated on the PhysioNet Non-EEG corpus of Birjandtalab et al. [19]: twenty subjects wearing an Affectiva Q wrist sensor for electrodermal activity and skin temperature, with pulse-oximeter heart rate, during relaxation and cognitive, emotional, and physical stress phases. Cognitive and emotional stress are compared with relaxation; the physical-exertion phase is held out because its elevated heart rate represents a different mechanism.

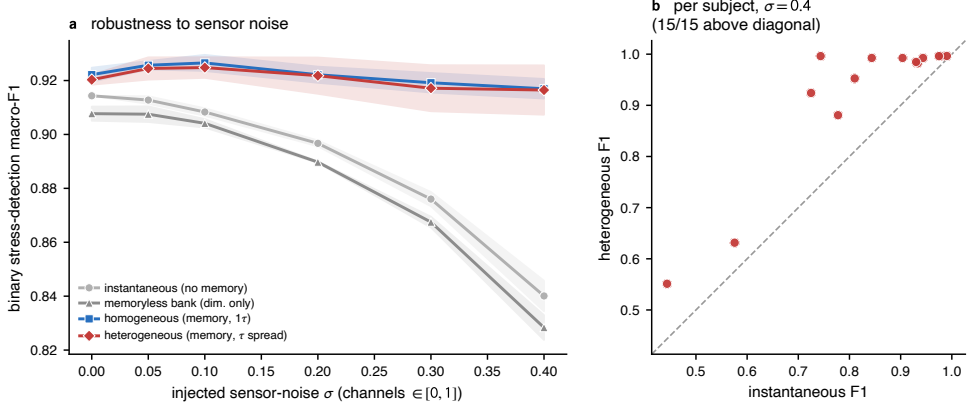


Figure 5.8: Continuous stress monitoring with device reservoirs (binary stress vs. baseline-plus-amusement, per-step, leave-one-subject-out, linear read-out). (a) Detection macro-F1 against the magnitude σ of injected sensor noise and motion-artefact bursts. Fading-memory banks remain near 0.92, while instantaneous and memoryless controls decrease with noise; bands are ± 1 SD over seeds. (b) At $\sigma = 0.4$, the heterogeneous memory bank exceeds the instantaneous read for all fifteen subjects (paired Wilcoxon $p < 10^{-4}$). Homogeneous and heterogeneous banks have similar scores, associating this result with fading memory rather than timescale spread.

With the same detector, clean macro-F1 is ≈ 0.65 – 0.68 , compared with ≈ 0.91 on WESAD, and the clean-stream memory difference is not resolved. Under injected noise, the instantaneous detector decreases from 0.651 to 0.516, while the fading-memory bank remains near 0.66. At $\sigma = 0.4$, the difference is $+0.130$, positive in nineteen of twenty subjects (paired Wilcoxon $p < 10^{-5}$; table 5.5). This independent corpus reproduces the noise-response ordering with a different sensor and population.

Table 5.5: Noise-response comparison on the independent PhysioNet Non-EEG stress corpus [19] (twenty subjects; Affectiva Q wrist electrodermal activity and temperature plus pulse-oximeter heart rate; cognitive and emotional stress versus relaxation; leave-one-subject-out, linear read-out). At $\sigma = 0.4$, the heterogeneous-minus-instantaneous difference is $+0.130$, positive in 19/20 subjects ($p < 10^{-5}$). Homogeneous and heterogeneous banks again have similar scores.

Detector	macro-F1 (clean)	macro-F1 ($\sigma = 0.4$)
Instantaneous (no memory)	0.651	0.516
Homogeneous (memory)	0.682	0.657
Heterogeneous (memory)	0.667	0.646

5.10 Toward a Worn Device: Sensor Site and Operating Budget

The robustness of the previous section is a laboratory statement until two further questions are answered: whether the advantage survives a consumer-grade sensor, and what the device would cost to run. Both are answered from the same measured ingredients (the WESAD wrist recordings and the device constants of ??) without any new assumption.

Sensor site. WESAD records each session with both a chest strap and an Empatica E4 wrist band. The wrist band has fewer and noisier channels: skin conductance, skin temperature, and photoplethysmography-derived heart rate, without respiration. On the wrist data (table 5.6), the clean macro-F1 of the instantaneous detector falls from 0.914 to 0.881, whereas the fading-memory reservoir changes from 0.920 to 0.922. The memory advantage is consequently larger on the wrist than on the chest ($+0.041$ versus $+0.006$ on clean streams and $+0.069$ under injected noise). Fading memory also reduces the wrist false-alarm rate from 0.062 to 0.040.

Table 5.6: Continuous binary stress-detection macro-F1 on the WESAD chest strap versus the Empatica E4 wrist band (leave-one-subject-out, linear read-out). Moving to the noisier consumer sensor degrades the instantaneous (static) detector but not the fading-memory reservoir, so the memory advantage is larger on the wrist; σ is the injected sensor-noise level of fig. 5.8.

Detector	Chest		Wrist E4	
	clean	$\sigma = 0.4$	clean	$\sigma = 0.4$
Instantaneous (no memory)	0.914	0.840	0.881	0.847
Memory reservoir (het)	0.920	0.916	0.922	0.916

Operating budget. The device dynamics are fixed, and only a linear read-out is fitted by one closed-form ridge solve. The trained component is an $(N+1) \times K$ weight matrix: 75 weights for the $N = 24$, three-class banks and 147 for the $N = 48$ banks. The deep models commonly used for wearable affect contain 10^4 – 10^6 trainable weights, two to four orders of magnitude more. Using the measured proof-of-concept energy of ≈ 50 nJ per write event (≈ 6 fJ μm^{-2} over a 0.0825 cm² junction), charging $N = 24$ nodes once per second corresponds to ≈ 1.2 μW . The linear read-out requires NK multiply–accumulates, well under a nanojoule per step. These estimates use unoptimised large-area devices; smaller junctions would reduce the event energy.

The estimate covers only the reservoir core: device charging and read-out operations. It excludes the sensor analogue front-end, analogue-to-digital conversion, and microcontroller overhead and is not a full system-power budget. The device response remains faster than the sub-hertz physiological signals considered here, and its relaxation performs the temporal filtering during the write/read cycle. The assumptions are tabulated in Appendix D.

5.11 Matching Device and Task Timescales

Performance depends on the overlap between device and task timescales (fig. 5.4). In the single-composition sweep, the lead cell gives the highest memory capacity for the seconds-scale benchmark (fig. 5.2). For the multi-lag physiological task, no single cell covers the full delay range, and the heterogeneous bank gives the highest single-bank score (table 5.2).

Effect of timescale breadth. Table 5.7 separates the effect of fading memory from that of timescale breadth. Both homogeneous and heterogeneous memory banks outperform memoryless and instantaneous baselines, with gains of up to +0.08 macro-F1 under sensor noise (sections 5.9 and 5.10). Heterogeneity provides an additional gain when several lags are required: memory capacity increases by a factor of 1.54 in simulation (positive in all ten seed pairs; fig. 5.1), and physiological-context reconstruction improves by +0.012 held-out R^2 over the best homogeneous bank (15/15 subjects, $p < 10^{-4}$; table 5.2). A multi- τ filter similarly improves continuous valence/arousal tracking by +0.176 CCC (27/30 subjects, $p < 10^{-6}$; section 5.8). The seed result is a sensitivity summary; the subject-level results support population inference. When one slow timescale dominates, as in the sustained stress labels and single-channel monitoring task, the best-matched homogeneous bank is as good as or marginally better than the heterogeneous bank. A composition-diverse bank is therefore useful for an unknown mixture of tasks: it loses at most about 0.01 macro-F1 on the single-timescale tasks considered here and gains more on the multi-timescale tasks. The available variation comes from composition and device-to-device scatter; chemistry and drive amplitude provide additional illustrative axes in ??.

Table 5.7: Comparison of instantaneous, homogeneous, and heterogeneous banks across the affective tasks in this chapter. Both memory banks exceed the instantaneous baseline, with the largest differences under sensor noise. The heterogeneous bank gives the highest physiological-context score, while the best-matched homogeneous bank gives the highest score for sustained labels and single-channel monitoring; homogeneous-heterogeneous differences on the classification tasks remain below about 0.01 macro-F1. Bold marks the highest score per row. Scores from tables 5.2 and 5.4 and sections 5.9 and 5.10.

Affective task	Metric	Instantaneous	Homogeneous	Heterogeneous
Physiological context (multi- τ)	R^2	0.673	0.744	0.756
Three-class affect tracking	macro-F1	0.709	0.753	0.758
Stress monitoring, clean	macro-F1	0.914	0.922	0.920
Stress monitoring, $\sigma=0.4$	macro-F1	0.840	0.917	0.916
Wrist stress, clean	macro-F1	0.881	0.930	0.922
Wrist stress, $\sigma=0.4$	macro-F1	0.847	0.918	0.916

Operating point and integration. The models apply in the measured sub-kilohertz regime, which covers the physiological signals used here but not fast logic. The assumed sub-threshold, state-preserving read sets the read-voltage and read-disturb limits for a circuit implementation. Figure 5.9 reports total memory capacity as device-to-device scatter is varied. The heterogeneous bank retains its advantage across the tested range, including the measured $\sigma(\ln \tau) \approx 0.85$. This value is estimated from silver PEO/LiOTf devices within each composition cell and is not explained by spin speed, anneal, film thickness, or device age (Appendix D). At zero scatter, a bank of identical nodes has little memory capacity because its states are redundant. A homogeneous bank therefore depends on within-composition scatter, whereas the heterogeneous bank also contains composition-dependent timescale differences.

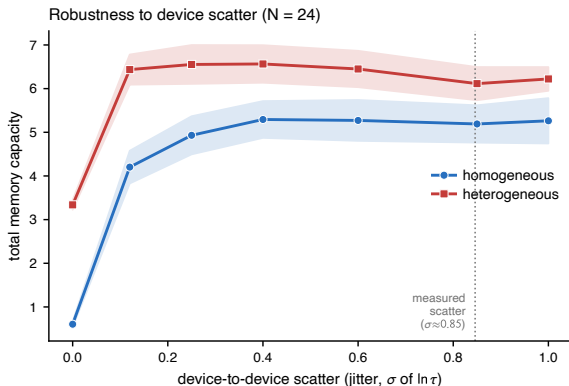


Figure 5.9: Total memory capacity versus injected device-to-device scatter (jitter on τ , β , and α) for homogeneous and heterogeneous banks ($N = 24$, ten seeds; shaded bands ± 1 SD). The heterogeneous bank retains a higher capacity across the tested range, including at the measured scatter (dotted line: $\sigma(\ln \tau) \approx 0.85$, derived in Appendix D). At zero scatter, identical node states are redundant and the homogeneous bank has little memory capacity.

5.12 Limitations and Scope Conditions

The results are subject to the following scope conditions.

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 5.2, not a bench reservoir, and every figure caption names the data behind it; the step from a fitted per-device model to a wired hardware reservoir is left to future work.

The composition assumption. As detailed in section 5.2, write non-linearity and fading memory were measured separately, at different write amplitudes and one inter-pulse cadence. Combining them assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. In a sensitivity analysis, each node’s retention was rescaled with its write strength across a wide coupling

range. The heterogeneous-to-homogeneous capacity ratio remained between 1.15 and 1.53 (Appendix D). The gold/TMPE negative-polarity measurements provide one empirical coupling estimate: write depth and decay half-time have a rank correlation of $\rho = +0.46$ across 28 single-run traces. Calibrating the sensitivity model to this value gives $\kappa^* = +0.18$, at which the capacity ratio is 1.45 and the heterogeneous bank is better in all ten paired random-seed runs (Appendix D). That corpus also shows interval-dependent potentiation, but its host, electrode, and polarity differ from the lead-cell model. A matched-amplitude experiment on the lead composition that varies pulse count and inter-pulse interval together would remove this assumption.

Operating regime and read-out. The model is identified only in the measured sub-kilohertz regime; fast temporal tasks are outside its scope. Every read-out is a single ridge regression because a nonlinear read-out could supply the task nonlinearity independently of the reservoir. The read is assumed to be sub-threshold and state-preserving.

Heterogeneity axes. The quantitative heterogeneous banks span the replicated PEO/LiOTf composition grid. The chemistry and drive-amplitude axes of ??, being sample-limited ($n \leq 2$ per matched cell), enter only as illustrative additional controls; Demonstration B is therefore a principle demonstration of composition heterogeneity, not a powered comparison across chemistries. The composition cells that define these banks index *composition*, not film thickness: although spin-coating speed and residual film thickness covary with the PEO fraction, the parameter cards inherited here were shown to be governed by composition, not thickness (??, ??), so the heterogeneity exploited in this chapter is compositional in origin.

Corpora and negative controls. The affective results use WESAD as the main corpus. Noise robustness is replicated on the PhysioNet Non-EEG dataset (section 5.9), and continuous valence and arousal are tested on CASE (section 5.8). A broader multi-corpus study remains future work. Two comparisons do not show a device-specific improvement: timescale heterogeneity does not separately improve final-label classification when one slow timescale dominates (section 5.7), and the nonlinear device encoding does not exceed a linear multi- τ filter on smooth continuous-affect regression (section 5.8). By contrast, the heterogeneous bank improves the multi-timescale memory-capacity and physiological-context tasks.

5.13 Summary

Behavioural models fitted to the measurements of ?? support temporal-computing simulations with a linear read-out. A heterogeneous bank raises memory capacity by a factor of 1.54 and improves multi-lag physiological-context reconstruction by +0.012 held-out R^2 over the best homogeneous bank. On WESAD, the models reach leave-one-subject-out macro-F1 scores of 0.894 for binary stress and 0.758 for three-class streaming classification. Under injected sensor noise, fading-memory banks remain near 0.92 macro-F1 while the instantaneous baseline falls to 0.84 (+0.088, all fifteen subjects, $p < 10^{-4}$). A classical moving average matches this robustness, attributing it to fading-memory filtering. The model’s nonlinearity provides no resolved increment on this task. The proposed material implementation would realise that filter at an estimated reservoir-core power of about a microwatt, with tens to hundreds of trained linear-read-out weights.

The additional value of a timescale distribution depends on the task. It is resolved for random-input memory, physiological-context recon-

struction, and continuous valence/arousal tracking, all of which depend on several lags. It is not resolved for the sustained WESAD labels or single-timescale stress monitoring. Composition-dependent relaxation and within-composition scatter provide the distributions used in these simulations.

The next experimental steps are a matched-amplitude pulse protocol that varies pulse count and cadence, a replicated study of chemistry and drive-amplitude heterogeneity, and a wired device bank that tests the behavioural-model predictions in hardware.

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